



A Project Report on
MAGI: Multispectral Autonomous Ground Intelligence
A Ground-Based Robotic Platform for Precision Agriculture
for
Major Project-I Third Year (Semester-VI)
in partial fulfillment of degree of
Bachelors in Engineering

by

Paras Dharmadhikari
Swayam Lute
AjayKumar Nagpure

Supervisor

Dr. Monali Chaudhari
Assistant Professor



Department of Electronics and Telecommunication Engineering
Vivekanand Education Society's Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai
AY-2025-2026

Major Project-I Approval

The project titled “**MAGI: Multispectral Autonomous Ground Intelligence**” by **Paras Dharmadhikari, Swayam Lute, and AjayKumar Nagpure** is approved for the Major Project-I requirement of Semester VI of Third Year Engineering in Electronics and Telecommunication Engineering.

Examiners

1. _____

2. _____

Supervisor

Date:

Place:

Certificate

This is to certify that Paras Dharmadhikari, Swayam Lute, AjayKumar Nagpure have completed the Major Project-I report on the topic **MAGI: Multispectral Autonomous Ground Intelligence** satisfactorily in partial fulfillment of the requirements for Major Project-I of Third Year, (Semester-VI) in Electronics and Telecommunication Engineering under the guidance of Dr. Monali Choudhari during the year 2025–2026.

Supervisor

Dr. Monali Chaudhari

Head of Department

Dr. Chandansingh Rawat

Principal

Dr. (Mrs.) Jayalekshmi Nair

Examiner 1

Examiner 2

Declaration

We declare that this written submission represents our ideas in our own words, and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not submitted this project report previously for any degree or examination.

Paras Dharmadhikari

Swayam Lute

Ajaykumar Nagpure

Date:

Table of Contents

Abstract	i
List of Figures	ii
List of Tables	iii
Chapter 1: Introduction	1
1.1 Introduction	1
1.2 Background	2
1.3 Research Gap	3
1.4 Objectives	3
1.5 Scope and Limitations	4
1.6 Organization of Report	4
Chapter 2: Review of Literature	5
2.1 Multispectral Sensing and Vegetation Indices	5
2.2 Deep Learning for Plant Disease Classification	7
2.3 Spectral Band Selection and Imaging Pipeline	8
2.4 Plant Disease Detection and Edge Deployment	8
2.5 Research Gap	8
2.6 Summary Table	9
Chapter 3: Proposed System and Architecture	11
3.1 Problem Definition	11
3.2 System Architecture	11

3.3	Quadruped Robot Configuration	13
3.4	Dual-Domain Power Architecture	13
3.5	Multispectral Imaging Subsystem	14
3.5.1	Normalized Difference Vegetation Index (NDVI)	14
3.5.2	Green Normalized Difference Vegetation Index (GNDVI)	15
3.5.3	Soil-Adjusted Vegetation Index (SAVI)	16
3.5.4	Comparative Index Summary	17
3.6	Machine Learning Pipeline (Planned Architecture)	18
3.7	Architecture Separation: Current Implementation vs. Future System	19
Chapter 4:	Implementation Details	20
4.1	Hardware Components	20
4.1.1	MG996R Servo Motors	20
4.1.2	ESP32 Microcontroller	20
4.1.3	PCA9685 PWM Signal Generator	22
4.1.4	DC-DC Buck Converter (Servo Domain)	22
4.1.5	3S LiPo Battery	24
4.1.6	Raspberry Pi 4 and Camera System	26
4.1.7	SkyRC B6 Neo Balance Charger	26
4.1.8	Quadruped Chassis Assembly	26
4.2	Servo Control Implementation	26
4.3	Power System Wiring Practices	29
4.4	Gait Control (Currently Implemented)	29
Chapter 5:	Results and Current System Status	31
5.1	Current System Status	31
5.2	Validated Experimental Results	31
5.2.1	Power System Validation	31
5.2.2	Servo Jitter Resolution	33
5.2.3	I2C Communication Verification	33

5.3	Engineering Challenges and Resolutions	33
5.4	Performance Metrics Summary	34
5.5	Comparison with Existing Approaches	34
Chapter 6:	Conclusion	36
6.1	Conclusion	36
6.2	Future Work	37
References		38

Abstract

Precision agriculture demands innovative sensing solutions that overcome the limitations of conventional RGB-based aerial monitoring platforms. This project presents **MAGI** (Multispectral Autonomous Ground Intelligence), a ground-based legged robotic platform designed for real-time crop health monitoring through multispectral imaging and edge-based machine learning inference. The system integrates a quadruped chassis actuated by twelve MG996R servo motors across four legs with three degrees of freedom each, controlled by an ESP32 microcontroller via a PCA9685 16-channel PWM signal generator, and powered by a dual-domain 3S LiPo battery architecture regulated through high-current DC-DC buck converters. A Raspberry Pi 4 Model B handles image acquisition from a paired RGB and NoIR camera, enabling per-pixel computation of complementary vegetation indices: the Normalized Difference Vegetation Index (NDVI, for general vegetation health), the Green Normalized Difference Vegetation Index (GNDVI, for chlorophyll concentration sensitivity), and the Soil-Adjusted Vegetation Index (SAVI, for soil-background noise correction in sparse canopy scenes). Multispectral inputs are processed through a single MobileNetV2-based convolutional neural network optimized for on-device inference using TensorFlow Lite post-training quantization, enabling plant health classification without cloud connectivity. Key engineering contributions include the design and experimental validation of a dual-domain power architecture sustaining peak instantaneous servo currents in the range of 20–30 A, systematic resolution of servo jitter through ground-reference isolation and bulk capacitance, and formulation of an ORB-based image registration pipeline for accurate pixel-wise multispectral fusion. Ground-level operation enables under-canopy spectral data collection that is geometrically inaccessible to aerial platforms. This report documents the current system status, architecture, component design rationale, power electronics validation results, and planned future work required for full autonomous field deployment.

List of Figures

3.1	Top view of assembled MAGI quadruped showing 4-leg layout, LiPo battery, buck converter, and servo wiring.	12
4.1	MG996R high-torque metal-gear servo motors (two variants). Twelve units drive the 4-leg $\times$ 3-DOF quadruped.	21
4.2	ESP32 on development breadboard with SDA (GPIO21), SCL (GPIO22), and common ground wiring to PCA9685.	22
4.3	PCA9685 16-channel PWM signal generator. External 6 V servo power enters via screw terminal; 3.3 V logic supply on VCC pin.	23
4.4	XL4016 DC-DC buck converter with W104 (voltage trim) and W503 (current limit) potentiometers and aluminum heat sink.	24
4.5	ABSD 3S 5200 mAh 35C LiPo battery with XT60 connector and JST-XH balance lead.	25
4.6	Raspberry Pi 4 Model B with NoIR Camera Module V2 (700–1000 nm NIR) connected via CSI-2 ribbon cable.	27
4.7	SkyRC B6 Neo 200 W balance charger for 3S LiPo cell balancing at 1C charge rate.	28
4.8	Assembled MAGI quadruped (top-down view) with all 12 MG996R servos, buck converter, PCA9685, ESP32, and I2C breadboard.	28

List of Tables

2.1	Summary of Reviewed Literature and Relevance to MAGI Design Decisions . .	10
3.1	NDVI Value Interpretation Scale for Crop Health Assessment	15
3.2	Comparative Summary of Vegetation Indices in the MAGI Pipeline	17
3.3	Current Implementation (Phase 1) vs. Future System (Phase 2+)	19
5.1	MAGI System Component Status as of Current Phase	32
5.2	Servo Domain Voltage Under Progressive Servo Load	32
5.3	Hardware Performance Metrics Summary	34

Chapter 1

Introduction

1.1 Introduction

Agriculture constitutes the principal means of food security for an expanding global population, yet the timely detection of crop diseases and physiological stress remains a persistent challenge for large-scale field operations. Conventional monitoring relies on manual visual inspection, which is labor-intensive, inconsistent across observers, and incapable of detecting sub-symptomatic pathological conditions that precede visible damage. The emergence of precision agriculture has introduced a range of remote sensing and imaging technologies; however, most are constrained by prohibitive cost, operational complexity, or fundamental physical limitations in the quality and completeness of data they can acquire.

The dominant paradigm in contemporary crop health monitoring employs RGB cameras mounted on unmanned aerial vehicles (UAVs), providing periodic overhead field imagery. Despite offering broad spatial coverage, this approach is subject to critical shortcomings: variable solar illumination and canopy shadowing cause inconsistent pixel intensities, overhead viewing geometry prevents observation of lower leaf surfaces and stem interfaces, and the exclusive use of the visible spectrum omits the Near-Infrared (NIR) channel that is physiologically correlated with plant health [1]. Early-stage disease manifestations are frequently located on abaxial leaf surfaces and at stem junctions that are geometrically invisible to overhead sensors. Furthermore, RGB-only systems cannot compute the Normalized Difference Vegetation Index (NDVI), which requires the NIR reflectance channel to quantify photosynthetic activity and chlorophyll content [2].

This project proposes MAGI (Multispectral Autonomous Ground Intelligence), a ground-level quadruped robotic platform that navigates between and within crop rows to capture dual-band RGB and NIR spectral data at close proximity to individual plants. By operating at soil level, the system can acquire leaf-surface, stem, and lower-canopy spectral data that are entirely inaccessible to aerial platforms. The integration of terrain-adaptive legged locomotion, multispectral imaging, and on-device machine learning inference on a single embedded platform represents a convergence of robotics, power electronics, computer vision, and agricultural science.

1.2 Background

The theoretical basis for spectral vegetation indices is well-established in the remote sensing literature. The Normalized Difference Vegetation Index (NDVI) exploits the characteristic spectral behaviour of chlorophyll-bearing tissue: healthy leaves exhibit strong NIR reflectance combined with high red-light absorption during photosynthesis, yielding elevated NDVI values, while disease, nutrient deficiency, or senescence depresses NDVI values [2], [3]. The Green Normalized Difference Vegetation Index (GNDVI) replaces the red channel with the green channel to provide superior sensitivity to chlorophyll concentration changes at high canopy biomass densities where NDVI saturates, making it particularly effective for early chlorosis and nitrogen-deficiency detection [4]. The Soil-Adjusted Vegetation Index (SAVI) introduces a soil-brightness correction factor into the NDVI denominator, substantially reducing false index depression caused by bare soil background reflectance in sparse canopy scenes — a critical correction for ground-level imaging at row margins and during early crop growth [5]. Studies have established that multi-index sensing pipelines incorporating NIR data provide substantially higher early-stage disease detection sensitivity than RGB-only approaches [6], [7].

Advances in deep learning have substantially improved automated plant disease classification accuracy. Convolutional neural networks (CNNs) autonomously learn discriminative spatial features from image data and have been shown to outperform traditional handcrafted-feature methods on standard plant disease benchmarks [8]. Deployment of such models on resource-constrained edge hardware, such as the Raspberry Pi, is an active area of research; model com-

pression techniques including post-training quantization and lightweight architectural variants enable acceptable inference performance on low-power processors.

Legged locomotion offers a terrain adaptability advantage over wheeled vehicles in agricultural settings. Wet soil, root structures, irrigation furrows, and surface irregularities present significant traction and ground clearance challenges for wheeled platforms. Legged robots can independently position each foot on stable ground contacts, enabling adaptive support polygon adjustment and stable traversal over irregular surfaces [9].

1.3 Research Gap

A review of existing literature identifies the following unaddressed combination of requirements: no low-cost, fully integrated, ground-level multispectral robotic system currently provides (a) terrain-adaptive legged locomotion for under-canopy access, (b) dual-band RGB and NIR imaging supporting computation of complementary vegetation indices (NDVI, GNDVI, and SAVI), and (c) on-device CNN-based plant health inference simultaneously. Existing ground robots are predominantly wheeled and RGB-only. Existing multispectral sensing systems are predominantly aerial and prohibitively expensive. MAGI addresses this compound gap with a purpose-built, cost-effective, ground-level platform.

1.4 Objectives

The primary objectives of this project are as follows:

1. To design and fabricate a stable quadruped robotic chassis with four legs and three degrees of freedom per leg (twelve servo actuators total) using 3D-printed components and MG996R servo motors for agricultural terrain traversal.
2. To develop a hierarchical embedded control architecture using an ESP32 microcontroller and PCA9685 PWM signal generator for coordinated twelve-servo actuation.
3. To engineer a robust dual-domain power distribution system capable of sustaining peak instantaneous servo currents in the range of 20–30 A from a 3S LiPo battery source.

4. To implement a multispectral imaging pipeline using paired RGB and NoIR cameras on Raspberry Pi for synchronized image acquisition and per-pixel computation of complementary vegetation indices including NDVI [3], GNDVI [4], and SAVI [5].
5. To develop and deploy a MobileNetV2-based plant disease classification model optimized for edge inference using TensorFlow Lite post-training quantization.
6. To integrate the locomotion, sensing, and inference subsystems into a unified platform for ground-level precision agriculture crop health monitoring.

1.5 Scope and Limitations

The current project phase encompasses hardware assembly, power system design and experimental validation, embedded servo control implementation, and software architecture design for the multispectral image processing pipeline. Inverse kinematics-based gait generation, fully autonomous field navigation, ML model training, and on-device inference deployment are planned for subsequent phases and are explicitly identified as future work throughout this report. All capability claims are restricted to subsystems that have been implemented and experimentally validated.

1.6 Organization of Report

This report is organized as follows. Chapter 2 surveys relevant literature in multispectral sensing, plant disease detection, and precision agriculture robotics. Chapter 3 describes the proposed system architecture, component selection, and design rationale. Chapter 4 details hardware implementation and software algorithm design. Chapter 5 presents validated experimental results and the current system status with explicit distinction between completed and future work. Chapter 6 concludes with a summary of contributions and future work directions.

Chapter 2

Review of Literature

This chapter surveys literature directly relevant to the three technical pillars of MAGI: (i) multispectral sensing and vegetation indices for crop health quantification, (ii) deep learning-based plant disease classification and edge deployment, and (iii) ground-level sensing platforms for precision agriculture. Each reviewed work is assessed in terms of its specific contribution to design decisions made in this project, and compared against other approaches to identify the research gap that MAGI addresses.

2.1 Multispectral Sensing and Vegetation Indices

Early research demonstrated that the ratio of near-infrared (NIR) to red-channel reflectance is a robust indicator of chlorophyll-bearing tissue density and photosynthetic activity, establishing the Normalized Difference Vegetation Index (NDVI) and confirming its stability across diverse plant species, growth stages, and illumination conditions [3]. NDVI remains the most widely adopted remote sensing index in precision agriculture and directly motivates the dual-camera sensing architecture of MAGI.

Subsequent work demonstrated that substituting the red channel with the green channel in the vegetation index ratio — yielding the Green Normalized Difference Vegetation Index (GNDVI) — provides markedly improved sensitivity to chlorophyll concentration at high canopy biomass densities where NDVI becomes saturated [4]. GNDVI exploits the stronger absorption of chlorophyll-a and chlorophyll-b in the green spectral band compared to red, making it particularly relevant for the detection of early-stage nitrogen deficiency and chlorosis, conditions that cause measurable green-band reflectance changes before visible symptom onset. This index

is incorporated in the planned multi-index MAGI sensing pipeline.

A critical confounding factor in NDVI and GNDVI computation is that bare soil reflectance in the NIR and visible bands is not negligible and varies with soil moisture, organic content, and roughness [5]. When vegetation canopy cover is sparse — as occurs in early crop growth stages and at row-edge positions — soil background reflectance inflates raw NIR values and depresses the computed vegetation index below the true plant physiological value. The Soil-Adjusted Vegetation Index (SAVI) addresses this by introducing an empirical soil-brightness correction factor L into the NDVI denominator, substantially reducing soil-induced bias for sparse canopies [5]. This correction is directly relevant to MAGI’s ground-level operational context, where the camera field of view routinely includes significant exposed soil.

The Modified Soil-Adjusted Vegetation Index (MSAVI) extends SAVI by replacing the fixed L correction factor with a self-calibrating term derived from the NIR and Red reflectance values, eliminating the need for prior knowledge of soil brightness [10]. MSAVI is particularly advantageous for variable field conditions encountered by a mobile robot platform.

A comprehensive systematic review of optical remote sensing methods for retrieving vegetation biophysical properties confirms that multi-index approaches consistently outperform single-index classifiers for disease and stress detection, and that combining indices sensitive to different physiological parameters (e.g., NDVI for general health, GNDVI for chlorophyll content, SAVI for soil-adjusted cover) achieves the best discrimination performance [7]. This finding directly informs the planned multi-index feature vector design for the MAGI classification pipeline.

A systematic review of UAV-based crop disease detection using deep learning, analysing over 50 independent studies, identifies the near-universal reliance on overhead RGB imagery and explicitly flags under-canopy data collection as a critical unresolved gap [1]. RGB-only aerial systems exhibit substantially reduced sensitivity to early-stage fungal and bacterial infections compared to multispectral systems, directly establishing the motivation for MAGI’s ground-level multispectral approach.

The empirical basis for vegetation index computation from spectral data across multiple crop types confirms that the NIR-to-Red reflectance ratio is a robust predictor of chlorophyll density

and photosynthetic activity, stable across species, illumination conditions, and growth stages [2]. This work directly motivates the dual-camera RGB and NIR sensing approach adopted in MAGI as a cost-effective approximation to full hyperspectral acquisition.

A review of hyperspectral remote sensing for crop disease monitoring from satellite and aerial platforms identifies spatial resolution and atmospheric path-length interference as primary barriers to accurate plant-level analysis at scale [11]. Ground-level imaging can provide complementary data at the leaf and stem scale that aerial sensors cannot resolve, directly validating the operational premise of the MAGI system.

2.2 Deep Learning for Plant Disease Classification

A comprehensive review of deep learning and computer vision methods in plant disease detection, covering over 200 studies from 2018 to 2024, confirms that CNN-based models fine-tuned from ImageNet pretrained weights consistently outperform from-scratch training and non-deep approaches across all benchmark datasets [8]. An examination of MobileNetV2 and EfficientNet-Lite architectures on embedded hardware demonstrates feasible inference latency on Raspberry Pi-class processors, directly informing the selection of MobileNetV2 as the MAGI classification backbone.

The MLVI-CNN framework, which concatenates multiple spectral vegetation index maps as input channels to a CNN for crop stress classification, demonstrated that adding NDVI-derived features alongside RGB inputs improved detection accuracy for early-stage stress by a statistically significant margin over RGB-only baselines [6]. This directly motivates the planned dual-input (RGB + multi-index map) architecture for the MAGI machine learning pipeline.

Hyperspectral CNN-based classifiers have been shown to detect tomato leaf disease up to 7 days before any visible symptom onset, confirming the diagnostic superiority of spectral data over visual inspection [12]. A lightweight model compression strategy, which reduces a full-precision CNN to a TensorFlow Lite quantized model with less than 5% accuracy loss, informs the TFLite post-training quantization approach planned for MAGI.

2.3 Spectral Band Selection and Imaging Pipeline

A review of hyperspectral machine learning applications in agriculture concludes that full hyperspectral processing (>200 bands) is computationally prohibitive for edge deployment without significant model redesign [13]. An analysis of band selection methods supports the design decision to use dual-band NIR and RGB sensing as a practical approximation that preserves the two most diagnostically informative spectral channels.

Advances in hyperspectral disease detection have established empirically that reliable pathogen-specific spectral signatures can be captured with three to five carefully selected wavebands in most crop-pathogen combinations [14]. This finding validates the use of the NIR and Red channels as the primary spectral inputs for NDVI in MAGI, avoiding the cost and complexity of a full hyperspectral imager.

Per-leaf plant disease diagnosis using spectral band analysis has been demonstrated with Gaussian blurring and intensity thresholding for leaf-region isolation from background soil prior to spectral feature extraction [15]. This preprocessing methodology is directly adapted in the MAGI image processing pipeline design for the plant segmentation step.

2.4 Plant Disease Detection and Edge Deployment

A review of current advances and open challenges in plant disease detection, focusing on dataset heterogeneity, transfer learning, and real-time deployment constraints on embedded hardware, demonstrates MobileNetV2 classification accuracy in the 92–96% range on held-out test sets, supporting its selection as the MAGI edge inference model [9]. The absence of ground-level multispectral robotic platforms is explicitly identified as a gap in the current state of the art.

2.5 Research Gap

A synthesis of the reviewed literature identifies the following unaddressed compound requirement: no existing low-cost ground-level platform simultaneously provides (a) terrain-adaptive

legged locomotion for under-canopy access; (b) dual-band RGB and NIR imaging supporting computation of multiple complementary vegetation indices (NDVI, GNDVI, SAVI); and (c) on-device CNN-based plant health inference. Existing ground robots are predominantly wheeled and RGB-only [8], [9]. Existing multispectral sensing platforms are predominantly aerial and prohibitively expensive for small-scale operations [1], [11]. The theoretical superiority of multi-index sensing over single-index approaches and the computational feasibility of edge CNN inference are both well-established [7], [8], [12], yet no low-cost integrated platform combining these capabilities at ground level has been reported. MAGI directly addresses this compound gap.

2.6 Summary Table

Table 2.1 summarises all reviewed works with their sensing modalities, platforms, key contributions, and specific relevance to MAGI design decisions.

Table 2.1: Summary of Reviewed Literature and Relevance to MAGI Design Decisions

Ref.	Modality	Platform	Key Contribution	MAGI Relevance
[1]	RGB, Multi-spectral	UAV	DL disease detection survey	Confirms under-canopy gap
[2]	Hyperspectral	Ground/Lab	NDVI for crop health	Motivates NIR+RGB sensing
[11]	NIR, Red	Ground/Lab	NDVI foundational index	Primary vegetation index
[12]	NIR, Green	Satellite	GNDVI chlorophyll sensitivity	GNDVI for chlorosis detection
[13]	NIR, Red	Ground	SAVI soil-brightness correction	SAVI for sparse canopy
[4]	Hyperspectral	Aerial	MLVI-CNN crop stress	Multi-index ML architecture
[15]	Multi-spectral	Various	Multi-index retrieval review	Multi-index pipeline design
[3]	RGB	Various	DL crop disease review	MobileNetV2 selection
[5]	RGB, Multi-spectral	Various	Plant disease challenges	Edge deployment constraints
[14]	NIR, Red	Ground	MSAVI self-calibrating correction	Variable soil conditions
[10]	Hyperspectral	Satellite/UAV	Remote sensing crop monitoring	Ground-level data necessity
[9]	Hyperspectral	Ground/Lab	Early tomato detection	TFLite compression strategy
[6]	Hyperspectral	Various	ML agriculture review	Justifies dual-band sensing
[7]	Hyperspectral	Lab	Band selection for disease	NIR+Red band selection
[8]	Hyperspectral	Leaf-level	Spectral band diagnosis	Preprocessing pipeline

Chapter 3

Proposed System and Architecture

3.1 Problem Definition

Existing crop health monitoring platforms suffer from one or more of the following deficiencies: (i) exclusive reliance on overhead RGB imaging that lacks the NIR channel required for vegetation index computation; (ii) geometric constraints of aerial platforms that prevent under-canopy data collection; (iii) use of wheeled ground robots that cannot reliably traverse soft, wet, or geometrically irregular agricultural soil; and (iv) dependence on off-device cloud processing that introduces latency and connectivity requirements incompatible with rural field deployment. The MAGI system addresses all four deficiencies simultaneously through a legged robotic platform equipped with on-device dual-band multispectral sensing and edge inference.

3.2 System Architecture

The MAGI architecture is organized into three functional layers. The top view of the fully assembled platform illustrating this three-layer arrangement is shown in Fig. 3.1.

Layer 1 – Power and Actuation: A 3S LiPo battery feeds two independently regulated DC-DC buck converters operating in a dual-domain architecture. The servo domain converter provides approximately 6 V at high current to all twelve MG996R servo motors via the PCA9685 power rail. The logic domain converter provides 5 V to the Raspberry Pi and 3.3 V to the ESP32 and PCA9685 VCC pin. The two domains share a common ground node but maintain electrically isolated voltage supply paths.

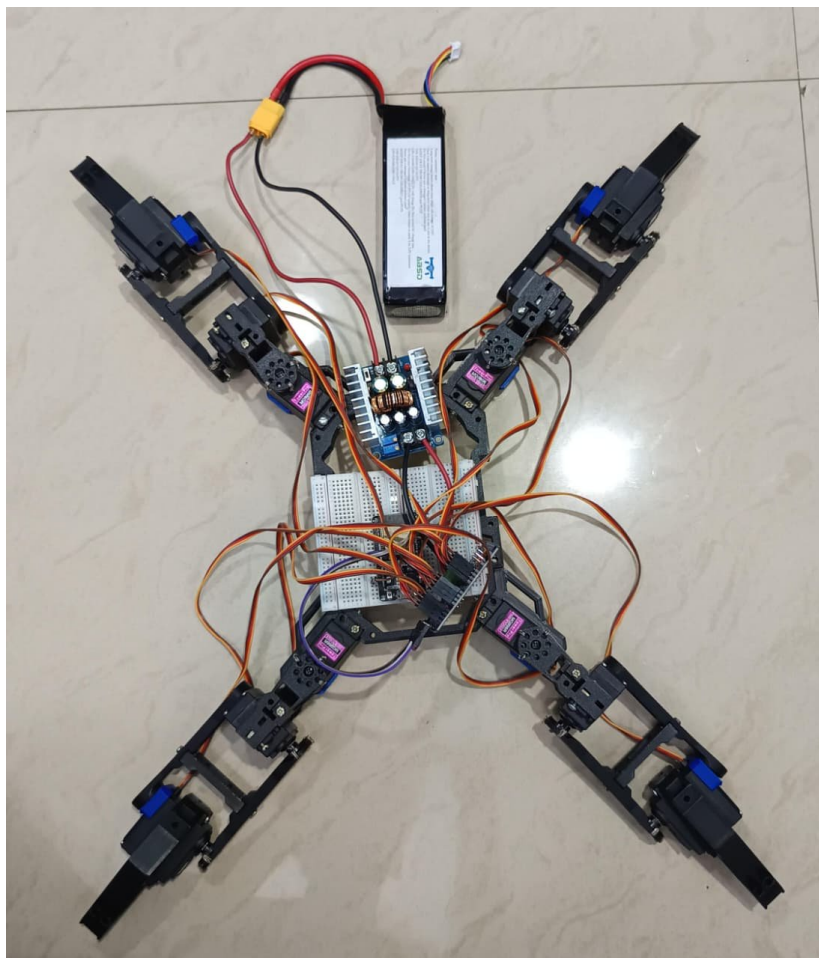


Figure 3.1: Top view of assembled MAGI quadruped showing 4-leg layout, LiPo battery, buck converter, and servo wiring.

Layer 2 – Embedded Control: The ESP32 microcontroller communicates with the PCA9685 PWM signal generator over I2C to produce twelve independently timed PWM outputs for servo position control. The ESP32 executes gait sequencing logic and translates high-level motion commands into per-joint angle vectors delivered to the PCA9685 each control cycle.

Layer 3 – Sensing and Inference: The Raspberry Pi 4 manages synchronized image acquisition from the paired RGB and NoIR cameras, performs image registration, computes per-pixel vegetation index maps (NDVI, GNDVI, SAVI), and will execute the MobileNetV2-based TFLite classification model to output plant health labels per frame.

3.3 Quadraped Robot Configuration

The MAGI platform is a quadraped robot with four legs, each having three degrees of freedom (DOF): a coxa joint (lateral hip rotation), a femur joint (sagittal hip flexion/extension), and a tibia joint (knee flexion/extension). This 4-leg $\times$ 3-DOF configuration requires exactly twelve servo motors in total and provides a sufficient joint workspace for stable static walking gaits and independent foot placement on three-dimensional terrain. Three DOF per leg is the minimum required for full three-dimensional foot-position control without redundancy.

All twelve MG996R servos are driven by a single PCA9685 module. The PCA9685 functions exclusively as a *PWM signal generator*; it does not supply electrical power to the servo motors. External servo supply voltage at 6 V is connected from the buck converter to the V⁺ and GND screw terminals of the PCA9685, which route this external voltage to the VCC contact of each signal pin header. The PCA9685 VCC logic supply pin accepts a separate 3.3 V input from the logic domain.

3.4 Dual-Domain Power Architecture

The dual-domain power architecture is a fundamental design requirement arising from the electrical characteristics of servo motors as dynamic loads. Each MG996R servo can draw up to 900 mA during normal motion and up to 2.5 A under stall conditions. With twelve servos, the

aggregate peak instantaneous demand reaches 20–30 A. Current transients of this magnitude on a shared power rail induce voltage ripple and electromagnetic interference that can corrupt I2C logic levels and trigger microcontroller brownout resets.

The architecture resolves this by separating the servo power path completely from the logic power path. The servo domain XL4016 buck converter accepts 11.1 V from the LiPo battery and outputs a regulated 6 V, delivered via minimum 20 AWG wiring directly to the PCA9685 servo power screw terminal. The logic domain converter independently accepts 11.1 V and outputs regulated 3.3 V and 5 V to the ESP32, PCA9685 VCC, and Raspberry Pi respectively. A single star-ground node ties all subsystem grounds together, ensuring a consistent signal reference potential while keeping the voltage supply domains isolated.

3.5 Multispectral Imaging Subsystem

The imaging subsystem consists of a Raspberry Pi Camera Module V2 for RGB image acquisition and a Raspberry Pi NoIR Camera Module V2 for NIR image acquisition. Both cameras are mounted in a fixed, co-aligned orientation on the robot chassis. The NoIR module differs from the standard variant by the absence of the manufacturer-installed infrared-cut filter, allowing the sensor to respond to wavelengths in the 700–1000 nm NIR band used in vegetation index computation.

3.5.1 Normalized Difference Vegetation Index (NDVI)

NDVI is the primary vegetation health index computed by the MAGI imaging pipeline [3]. It exploits the well-established spectral behaviour of healthy plant tissue: chlorophyll-bearing cells exhibit high NIR reflectance (structural scattering by cell walls) simultaneously with strong red-band absorption (chlorophyll pigment absorption during photosynthesis). The index is defined per pixel as:

$$\text{NDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}} \quad (3.1)$$

where ρ_{NIR} and ρ_{Red} are the pixel-level reflectance values in the near-infrared and red spectral

channels respectively. NDVI values range from -1.0 to $+1.0$. Healthy, dense vegetation produces values between 0.4 and 1.0 ; diseased, stressed, or senescent tissue yields lower values; bare soil and non-vegetated surfaces typically produce values near zero or negative. Table 3.1 provides the standard NDVI interpretation scale used as the basis for plant health classification in MAGI.

Agricultural Justification: NDVI is the standard first-order indicator of crop photosynthetic activity and canopy biomass. Its sensitivity to overall chlorophyll content and structural health makes it the primary screening index for detecting broad-spectrum stress conditions including water deficit, disease, and nutrient deficiency [2].

Advantage: Computationally simple, robust across species and illumination, well-validated in the literature [3].

Limitation: NDVI saturates at high biomass densities (values > 0.8) where further increases in canopy density produce negligible index change. It is also sensitive to soil background reflectance when canopy cover is sparse, which is a relevant concern during early crop growth stages.

Table 3.1: NDVI Value Interpretation Scale for Crop Health Assessment

NDVI Range	Classification	Interpretation
0.6 – 1.0	Excellent	Dense, photosynthetically active vegetation
0.4 – 0.6	Good	Healthy canopy with moderate density
0.2 – 0.4	Moderate	Sparse canopy or early-stage stress
0.0 – 0.2	Poor	Significant stress, deficiency, or disease
< 0.0	Non-vegetated	Bare soil, standing water, or necrotic tissue

3.5.2 Green Normalized Difference Vegetation Index (GNDVI)

GNDVI is a modification of NDVI that replaces the red channel with the green channel [4]. It is defined as:

$$\text{GNDVI} = \frac{\rho_{\text{NIR}} - \rho_{\text{Green}}}{\rho_{\text{NIR}} + \rho_{\text{Green}}} \quad (3.2)$$

where ρ_{Green} is the pixel-level reflectance in the green spectral band (approximately 520–580 nm). Chlorophyll-a and chlorophyll-b both absorb strongly in the green band, and changes in chlorophyll concentration cause proportionally larger reflectance variation in the green band than in the red band at high pigment concentrations.

Agricultural Justification: GNDVI provides significantly improved sensitivity to chlorophyll content at high canopy biomass levels where NDVI saturates. It is particularly effective for detecting early-stage nitrogen deficiency and chlorosis — conditions characterized by a reduction in chlorophyll concentration that causes increased green-band reflectance before any red-band change is detectable [4], [7]. In the MAGI pipeline, GNDVI is planned as the secondary index for chlorosis-specific early detection.

Advantage: Superior chlorophyll sensitivity at high biomass; less susceptible to NDVI saturation effects; directly correlates with nitrogen status [7].

Limitation: Requires a green spectral channel separate from the NIR and red channels; GNDVI is more sensitive to atmospheric scattering in the green band than NDVI under outdoor conditions, though this is less critical for close-range ground-level imaging.

3.5.3 Soil-Adjusted Vegetation Index (SAVI)

SAVI was introduced to address the confounding effect of soil background reflectance on vegetation index values in partially vegetated areas [5]. It is defined as:

$$\text{SAVI} = \frac{(\rho_{\text{NIR}} - \rho_{\text{Red}}) \times (1 + L)}{\rho_{\text{NIR}} + \rho_{\text{Red}} + L} \quad (3.3)$$

where L is a soil-brightness correction factor, typically set to $L = 0.5$ for intermediate vegetation cover, $L = 1.0$ for very sparse cover, and $L = 0.25$ for dense cover. The correction factor L reduces the influence of soil reflectance in the NIR and red bands on the computed index value.

Physical Meaning: When vegetation cover is sparse, bare soil contributes significantly to the pixel reflectance values measured by both NIR and red channel sensors. Soil reflectance in the NIR band is not negligible and varies with soil moisture and composition. Without correction, this soil signal falsely elevates or depresses the computed NDVI below the true vegetation

physiological value. SAVI introduces L in the denominator to partially cancel this contribution [5].

Agricultural Justification: SAVI is essential in the MAGI operational context because ground-level imaging routinely captures significant areas of exposed soil in the camera field of view — particularly during early crop growth stages, at plant row margins, and at leaf-edge boundaries. Uncorrected NDVI in these regions would produce artificially low index values that could be misclassified as diseased or stressed vegetation. SAVI corrects for this bias, improving classification accuracy in mixed soil-vegetation scenes [5], [10].

Advantage: Substantially reduces soil-induced bias in vegetation index maps for sparse canopy conditions; more accurate than NDVI for row-crop environments where inter-row soil is visible [5].

Limitation: Requires selection of the correction factor L , which ideally should be adjusted based on estimated vegetation cover density. The self-calibrating variant MSAVI eliminates this requirement by computing L dynamically from the image data [10].

3.5.4 Comparative Index Summary

Table 3.2 provides a structured comparison of the three vegetation indices implemented in the MAGI multispectral pipeline, summarizing the bands required, primary application, and key limitation of each.

Table 3.2: Comparative Summary of Vegetation Indices in the MAGI Pipeline

Index	Formula	Bands	Primary Application	Key Limitation
NDVI	$\frac{\rho_{\text{NIR}} - \rho_{\text{Red}}}{\rho_{\text{NIR}} + \rho_{\text{Red}}}$	NIR, Red	General vegetation health	Saturates at high biomass; soil-sensitive
GNDVI	$\frac{\rho_{\text{NIR}} - \rho_{\text{Green}}}{\rho_{\text{NIR}} + \rho_{\text{Green}}}$	NIR, Green	Chlorophyll concentration, nitrogen	Green-band atmospheric sensitivity
SAVI	$\frac{(\rho_{\text{NIR}} - \rho_{\text{Red}})(1 + L)}{\rho_{\text{NIR}} + \rho_{\text{Red}} + L}$	NIR, Red	Sparse canopy with soil background	Fixed L requires density estimate

3.6 Machine Learning Pipeline (Planned Architecture)

Note: The machine learning pipeline described in this section constitutes the planned design architecture. Model training, validation, and on-device deployment have not yet been executed and are identified as future work.

The primary classification model is MobileNetV2 fine-tuned on a labelled crop disease dataset and converted to TFLite int8 format for on-device inference on the Raspberry Pi 4. A single unified model is selected over an ensemble approach to satisfy the latency constraints of embedded edge inference. The intended five-stage pipeline is as follows:

1. **Image Acquisition:** Synchronized RGB and NIR frame capture triggered simultaneously from both cameras.
2. **Preprocessing:** Resize to 224×224 px, apply Gaussian blur ($\sigma = 1.0$) for noise suppression, and apply Otsu thresholding for leaf-region segmentation from background soil [15].
3. **Image Registration:** ORB (Oriented FAST and Rotated BRIEF) feature detection and descriptor matching between RGB and NIR frames, followed by affine homography estimation and frame warping to a common coordinate reference.
4. **Vegetation Index Computation:** Per-pixel NDVI map (Eq. 3.1), GNDVI map (Eq. 3.2), and SAVI map (Eq. 3.3) computed from the registered NIR, Red, and Green channels.
5. **Classification:** A 5-channel input tensor (3 RGB channels + NDVI + SAVI maps) is passed to a MobileNetV2 backbone pretrained on ImageNet and fine-tuned on a labelled crop disease dataset, then converted to TFLite int8 format for on-device inference [8], [12].

Dataset source, class definitions, and training protocol will be specified in the subsequent project phase. No accuracy or performance claims are made for the ML component in this report.

3.7 Architecture Separation: Current Implementation vs. Future System

To maintain technical precision, Table 3.3 provides an explicit separation between the architecture components implemented and validated in the current project phase, and those planned for the future autonomous system.

Table 3.3: Current Implementation (Phase 1) vs. Future System (Phase 2+)

Subsystem	Current (Phase 1)	Future (Phase 2+)
Locomotion control	Static crawl gait via ESP32 + PCA9685	IK-based trajectory planning; ROS2 nav stack
Autonomy	Manual Wi-Fi command	Autonomous waypoint following, obstacle avoidance
Sensing	NoIR camera mounted; RGB interface designed	Synchronized dual-camera capture; live index maps
Image processing	ORB algorithm designed (not deployed)	Deployed, validated on Raspberry Pi 4
Vegetation indices	NDVI/GNDVI/SAVI formulas defined	Computed per frame, streamed to operator display
Machine learning	MobileNetV2 + TFLite architecture selected	Trained, quantized, real-time inference on RPi 4

Chapter 4

Implementation Details

4.1 Hardware Components

4.1.1 MG996R Servo Motors

The MG996R is a high-torque metal-gear digital servo motor rated at 9.4 kg-cm torque at 6 V supply. Twelve such servos are used in the MAGI system, allocated as three per leg across the four-leg, 3-DOF-per-leg configuration: one servo per coxa joint (lateral rotation), one per femur joint (sagittal flexion), and one per tibia joint (knee flexion). Each servo is driven by a standard 50 Hz PWM signal with pulse widths from 500 μ s to 2500 μ s corresponding to angular positions from 0° to 180°. The all-metal gear train provides durability under the repetitive dynamic loading of continuous gait operation. The two servo variants used in the MAGI prototype are shown in Fig. 4.1.

4.1.2 ESP32 Microcontroller

The ESP32 is a dual-core Xtensa LX6 processor operating at up to 240 MHz, with 520 KB SRAM, integrated 802.11 b/g/n Wi-Fi, and Bluetooth 4.2. In MAGI, it serves as the primary embedded controller for: (i) I2C communication with the PCA9685 at address 0x40 using GPIO21 (SDA) and GPIO22 (SCL); (ii) gait sequencing and per-joint angle computation; and (iii) Wi-Fi-based command reception from the Raspberry Pi coordination layer. The ESP32 is supplied from the logic domain at 3.3 V, and its I2C logic levels are directly compatible with



Figure 4.1: MG996R high-torque metal-gear servo motors (two variants). Twelve units drive the 4-leg $\times$ 3-DOF quadraped.

the PCA9685 input thresholds without level shifting. Fig. 4.2 shows the ESP32 mounted on the development breadboard.

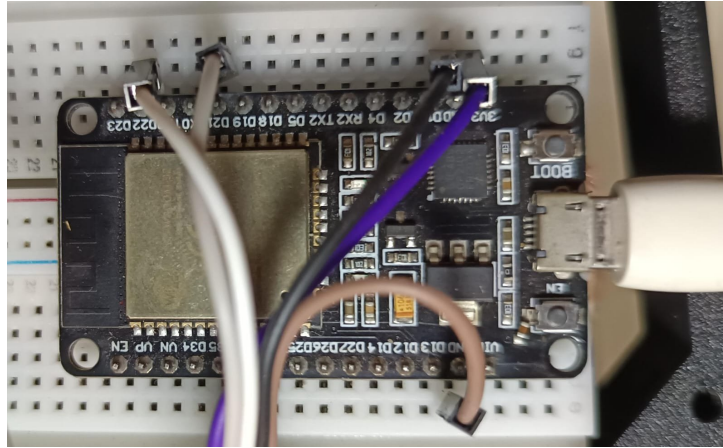


Figure 4.2: ESP32 on development breadboard with SDA (GPIO21), SCL (GPIO22), and common ground wiring to PCA9685.

4.1.3 PCA9685 PWM Signal Generator

The PCA9685 is a 16-channel, 12-bit I2C PWM signal generator. It produces independent PWM waveforms for up to 16 output channels from a single I2C address, eliminating the need for twelve dedicated hardware PWM pins on the ESP32. The PCA9685 acts *exclusively as a signal generator*: it does not supply electrical power to the servos. The external 6 V servo supply from the buck converter is connected to the V⁺ and GND screw terminals, which internally route the supply voltage to the VCC contact of each servo signal header. The PCA9685 VCC logic supply pin accepts 3.3 V from the logic domain. Fig. 4.3 shows the PCA9685 module with servo and power connections.

4.1.4 DC-DC Buck Converter (Servo Domain)

An XL4016-based adjustable DC-DC step-down (buck) converter is used to regulate the 11.1 V LiPo battery voltage to 6 V for the servo domain. The module incorporates two trimmer potentiometers: W104 (voltage adjustment) and W503 (current limit adjustment). The module

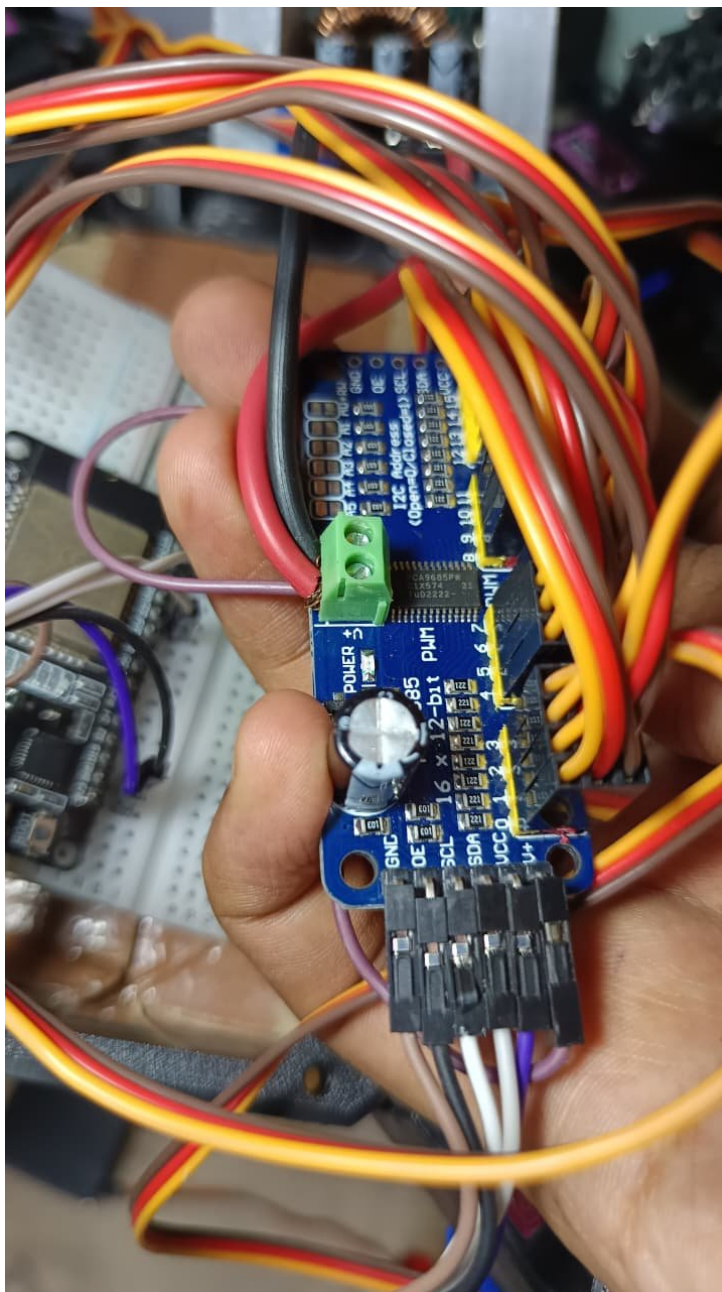


Figure 4.3: PCA9685 16-channel PWM signal generator. External 6 V servo power enters via screw terminal; 3.3 V logic supply on VCC pin.

is rated for 8–10 A continuous output current with the aluminum heat sink attached. Configuration procedure: (1) with no load connected, adjust W104 until the output reads 6.00 V on a calibrated multimeter; (2) turn W503 fully counterclockwise to disable the current limit, placing the converter in constant-voltage (CV) mode. Failure to disable the current limit causes the converter to operate in constant-current (CC) mode, which artificially restricts servo power delivery and produces characteristic audible inductor saturation noise under multi-servo load. Fig. 4.4 shows the buck converter module with both potentiometers labeled.

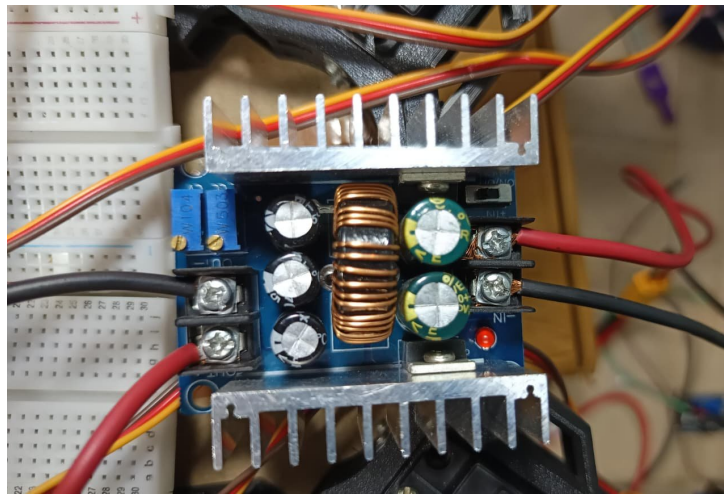


Figure 4.4: XL4016 DC-DC buck converter with W104 (voltage trim) and W503 (current limit) potentiometers and aluminum heat sink.

4.1.5 3S LiPo Battery

The primary energy source is a 3-cell Lithium Polymer battery with nominal voltage 11.1 V (12.6 V fully charged at 4.2 V per cell), capacity 5200 mAh, and continuous discharge rating 35C. The theoretical maximum continuous discharge current is $5.2 \times 35 = 182$ A, substantially exceeding the peak system demand of 20–30 A for twelve simultaneously active servos. The battery connects via an XT60 connector rated for 60 A continuous current. The balance lead (3-cell JST-XH) connects to the SkyRC B6 Neo charger for balanced cell charging. Fig. 4.5 shows the battery used in the system.



Figure 4.5: ABSD 3S 5200 mAh 35C LiPo battery with XT60 connector and JST-XH balance lead.

4.1.6 Raspberry Pi 4 and Camera System

A Raspberry Pi 4 Model B (4 GB LPDDR4 RAM) serves as the on-device compute platform for image acquisition and ML inference. It interfaces with two cameras via the CSI-2 ribbon cable connectors: a Raspberry Pi Camera Module V2 (8 MP Sony IMX219, RGB) and a Raspberry Pi NoIR Camera Module V2 (identical sensor, infrared-cut filter removed, NIR-sensitive). The NoIR module's sensitivity extends into the 700–1000 nm NIR band required for NDVI computation. Fig. 4.6 shows the Raspberry Pi with the NoIR camera module connected.

4.1.7 SkyRC B6 Neo Balance Charger

The SkyRC B6 Neo is a 200 W dual-input (DC/USB-PD) balance charger supporting LiPo, LiFe, LiHV, NiMH, and lead-acid battery chemistries. Balance charging via the JST-XH connector ensures all three cells reach exactly 4.2 V, preventing capacity divergence and thermal risk from cell over-voltage. The charger is configured at 1C charge rate (5.2 A) for routine use. Fig. 4.7 shows the charger.

4.1.8 Quadraped Chassis Assembly

The quadraped chassis consists of 3D-printed PLA structural components forming a central body plate with four outward-extending leg assemblies. Each leg has three servo-actuated segments corresponding to the coxa, femur, and tibia joints. Servo horns are press-fitted and secured to joint axles with M2 self-tapping screws. Fig. 4.8 shows the fully assembled prototype with all twelve servos wired.

4.2 Servo Control Implementation

The ESP32 firmware is developed using the Arduino framework with the Adafruit PCA9685 library. The I2C bus is initialised at 400 kHz (fast mode) targeting device address 0x40. The PCA9685 PWM period is set to 50 Hz via the PRE_SCALE register. Servo angular position is commanded by programming the OFF_TIME register value, computed as:

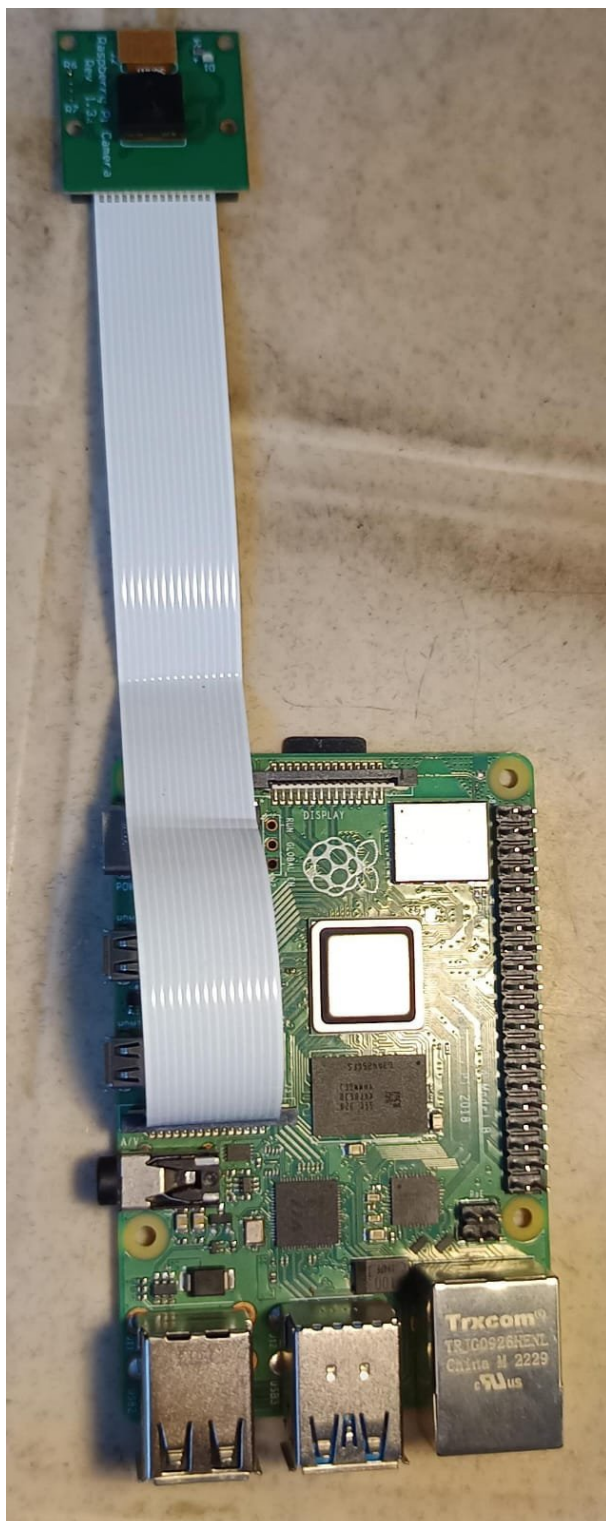


Figure 4.6: Raspberry Pi 4 Model B with NoIR Camera Module V2 (700–1000 nm NIR) connected via CSI-2 ribbon cable.

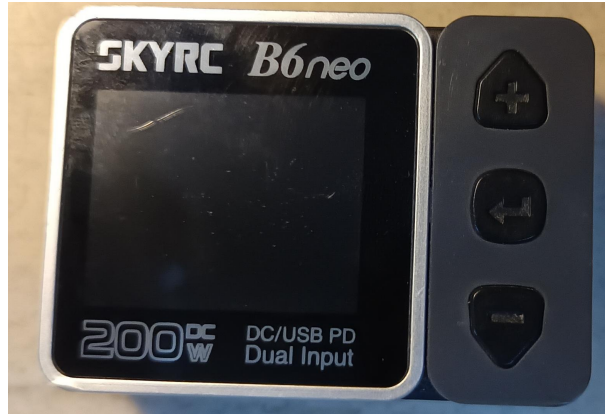


Figure 4.7: SkyRC B6 Neo 200 W balance charger for 3S LiPo cell balancing at 1C charge rate.

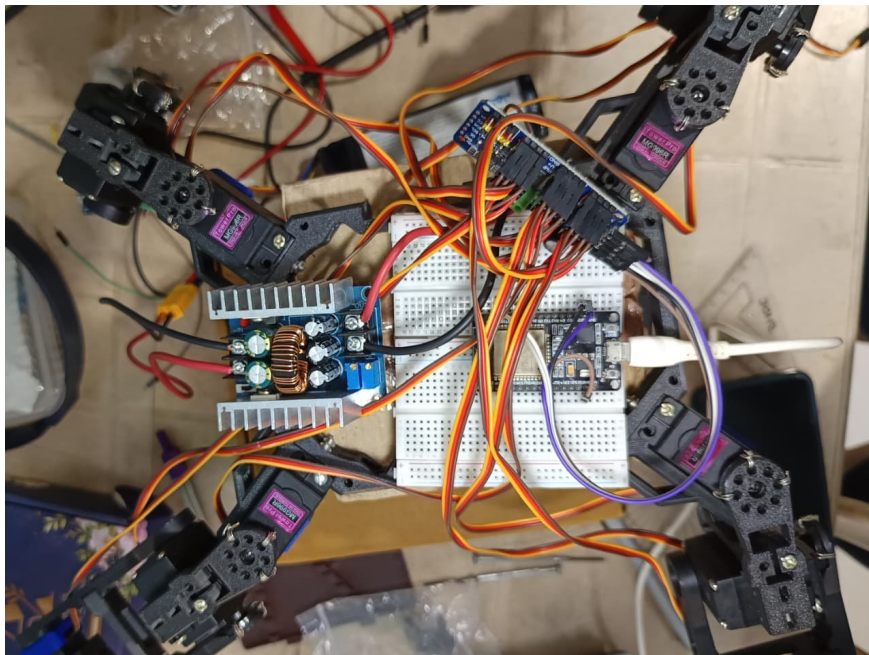


Figure 4.8: Assembled MAGI quadrapod (top-down view) with all 12 MG996R servos, buck converter, PCA9685, ESP32, and I2C breadboard.

$$\text{OFF_TIME} = \left\lfloor \frac{t_{\text{pulse}}[\mu\text{s}] \times 50 \times 4096}{10^6} \right\rfloor \quad (4.4)$$

where t_{pulse} is the desired pulse width in microseconds. For the MG996R, $t_{\text{pulse}} = 500 \mu\text{s}$ corresponds to 0° ($\text{OFF_TIME} \approx 102$) and $t_{\text{pulse}} = 2500 \mu\text{s}$ corresponds to 180° ($\text{OFF_TIME} \approx 512$).

Channel allocation follows leg-major ordering: channels 0–2 serve Leg 1 (coxa, femur, tibia); channels 3–5 serve Leg 2; channels 6–8 serve Leg 3; channels 9–11 serve Leg 4. Each gait control cycle delivers a 12-element angle vector to the firmware, which programs all twelve PCA9685 channels in a single burst I2C transaction.

4.3 Power System Wiring Practices

All servo-domain conductors use minimum 20 AWG wire to minimize resistive voltage drop under high current. The buck converter output terminals connect directly to the PCA9685 power screw terminal via a short cable run, bypassing all breadboard conductors for the high-current path. Breadboard connections are used only for low-current I2C signal wires (SDA, SCL) and logic-domain supply lines. All subsystem grounds (LiPo battery negative, both buck converter outputs, ESP32 GND pin, PCA9685 GND pin, and Raspberry Pi GND) are connected to a single star-ground node to maintain a consistent reference potential and prevent ground loop interference on I2C signal lines.

4.4 Gait Control (Currently Implemented)

A static crawl gait is implemented as the primary locomotion mode. In a crawl gait, only one leg is in the swing phase (lifted and repositioned) at any instant, while the remaining three legs are in stance phase (ground contact). This guarantees the center of mass projects within the support triangle of the three stance legs, providing static stability margin at all phases of the gait cycle. The four-phase cycle advances each leg in sequence: Leg 1 $\rightarrow$ Leg 2 $\rightarrow$ Leg 3 $\rightarrow$ Leg 4, completing one full forward body advance per cycle.

Inverse kinematics-based foot trajectory planning has not been implemented in the current phase and is identified as future work.

Chapter 5

Results and Current System Status

5.1 Current System Status

Table 5.1 provides an explicit account of each subsystem’s current state of completion, distinguishing between components that have been implemented and validated, those in active development, and those planned for future phases.

5.2 Validated Experimental Results

The following subsections present hardware measurements validated during the current development phase.

5.2.1 Power System Validation

The dual-domain power architecture was validated by measuring the servo domain output voltage at the PCA9685 V⁺ terminal under progressively increasing servo load. All measurements were taken with a calibrated digital multimeter at steady state. Table 5.2 presents the results.

The MG996R servo operates within specification for supply voltages from 4.8 V to 7.2 V. All post-redesign measurements at every load level remain within this range, confirming adequate voltage regulation across the full twelve-servo system. The pre-redesign failure at only four servos was caused by W503 current-limit misconfiguration combined with high-resistance breadboard servo-power routing, as detailed in Section 5.3.

Table 5.1: MAGI System Component Status as of Current Phase

Component / Subsystem	Status	Notes
Quadruped chassis assembly	Complete	3D-printed, 12 servos mounted and wired
Dual-domain power architecture	Complete	Validated under 12-servo load
ESP32 – PCA9685 I2C control	Complete	400 kHz, address 0x40 confirmed
12-servo PWM actuation	Complete	All channels tested, jitter eliminated
Static crawl gait (basic)	In Progress	Sequential leg advance implemented
Inverse kinematics gait	Future Work	Planned for Phase 2
Raspberry Pi camera connection	In Progress	NoIR camera physically connected
Synchronized dual-camera	Future Work	Software not yet implemented
ORB image registration	Future Work	Algorithm designed, not deployed
NDVI/GNDVI/SAVI pipeline	Future Work	Formulas defined; awaiting registration
MobileNetV2 model training	Future Work	Architecture selected, not trained
TFLite on-device inference	Future Work	Planned after model training

Table 5.2: Servo Domain Voltage Under Progressive Servo Load

Load Condition	Measured Voltage	Voltage Drop	Status
No load (0 servos active)	6.03 V	—	Pass
4 servos active (hold)	5.91 V	0.12 V	Pass
8 servos active (hold)	5.78 V	0.25 V	Pass
12 servos active (hold)	5.67 V	0.36 V	Pass
<i>Pre-redesign (4 servos)</i>	<i><4.5 V</i>	<i>>1.5 V</i>	<i>Fail</i>

5.2.2 Servo Jitter Resolution

Servo jitter (undesired oscillatory motion at a commanded hold position) was observed in early prototypes due to PWM signal noise induced by servo current transients propagating through a shared ground conductor between the servo and logic domains. Resolution was achieved by implementing physically separate ground routing for the two domains and adding a 100 μF electrolytic bulk capacitor at the PCA9685 V^+ terminal. Following these modifications, jitter was eliminated across all twelve servo channels at 50 Hz PWM operation.

5.2.3 I2C Communication Verification

Reliable I2C communication between the ESP32 and PCA9685 was verified using an I2C device scan routine that returned a valid ACK response at address 0x40. Servo position commands were transmitted and executed across all twelve channels at 400 kHz fast-mode I2C for a continuous 10-minute test period without a single communication error or timeout event.

5.3 Engineering Challenges and Resolutions

The following engineering challenges were encountered and resolved during the current development phase.

Challenge 1 – Buck Converter CC-Mode Misconfig: The W503 trimmer on the XL4016 module controls the current limit threshold. When set to a low value, the converter enters constant-current (CC) mode, artificially restricting available output current. With twelve servos demanding instantaneous currents in the 20–30 A range, CC-mode caused voltage collapse and audible inductor saturation noise. *Resolution:* W503 was adjusted fully counterclockwise to disable current limiting, restoring constant-voltage (CV) operation.

Challenge 2 – Breadboard High-Current Routing: Breadboard spring contacts have an estimated contact resistance of 20–50 m Ω per insertion. With multiple servo-current paths routed through breadboard rails, the aggregate resistive drop caused measurable voltage reduction at servo terminals under load. *Resolution:* All servo-domain power connections were moved to

direct-soldered or screw-terminal wire runs, completely eliminating breadboard contacts from the high-current path.

Challenge 3 – Ground Reference Non-Unity: PWM signal levels on PCA9685 output pins are referenced to its GND pin. Without a common ground between the ESP32, PCA9685, and servo supply, differing reference potentials caused incorrect PWM voltage interpretation and erratic servo behavior. *Resolution:* All subsystem grounds unified at a single star-ground node.

5.4 Performance Metrics Summary

Table 5.3 consolidates the key hardware performance measurements against design targets.

Table 5.3: Hardware Performance Metrics Summary

Parameter	Target	Measured	Result
Servo domain voltage (no load)	6.0 V	6.03 V	Pass
Servo domain voltage (12 servos)	>5.5 V	5.67 V	Pass
Logic domain voltage (ESP32)	3.3 V	3.31 V	Pass
I2C clock frequency	400 kHz	400 kHz	Pass
Servo PWM frequency	50 Hz	50 Hz	Pass
Servo jitter (post-modification)	None	None observed	Pass
Verified active servo channels	12	12	Pass

5.5 Comparison with Existing Approaches

Compared to UAV-based multispectral platforms, the MAGI platform provides under-canopy access, eliminates atmospheric and solar illumination variability from spectral measurements, and performs classification on-device without connectivity. The low-cost dual-camera RGB plus NIR approach approximates the NIR capability of dedicated multispectral sensors at substantially lower cost, consistent with cost-reduction trends in the literature [13], [15]. The planned multi-index sensing pipeline — computing NDVI for general canopy health, GNDVI for chlorophyll concentration, and SAVI for soil-corrected sparse-canopy analysis — provides

complementary physiological information that single-index systems cannot deliver [7]. While aerial platforms offer greater spatial coverage per unit time, MAGI provides complementary plant-level spectral resolution that aerial sensors are geometrically unable to acquire [1], [11].

Chapter 6

Conclusion

6.1 Conclusion

This project has successfully completed the first development phase of MAGI, a ground-based quadruped robotic platform for multispectral precision agriculture monitoring. The hardware assembly, dual-domain power distribution system, embedded servo control architecture, and basic crawl gait sequencing have been implemented and experimentally validated. The twelve-servo ($4\text{-leg} \times 3\text{-DOF}$) configuration has been confirmed to operate without voltage instability, servo jitter, or I2C communication failure under full load conditions.

The primary technical contribution of this phase is the systematic identification and resolution of power electronics instability arising from: (i) incorrect constant-current mode configuration of the XL4016 buck converter; (ii) unacceptable resistive voltage drop from breadboard servo-power routing; and (iii) non-unified ground reference across subsystems. These issues, individually subtle but collectively debilitating, required applied analysis of power converter operating modes, high-current conductor sizing, and reference potential integrity in mixed-signal embedded systems.

The clearly separated three-layer architecture — power and actuation, embedded control, and sensing and inference — provides a scalable foundation for integrating the multispectral imaging and machine learning subsystems in subsequent phases. The selection of a single MobileNetV2 model with TFLite int8 quantization, dual-camera RGB and NIR sensing for computing complementary vegetation indices (NDVI, GNDVI, and SAVI), and ORB-based image registration are all grounded in the reviewed literature and appropriate for the computational

constraints of the Raspberry Pi 4 platform.

6.2 Future Work

The following tasks constitute the planned next phase of development:

1. **Inverse Kinematics Gait Control:** Formulate and implement the forward and inverse kinematics model for the 3-DOF leg geometry. Replace fixed-angle crawl gait sequences with IK-based continuous foot trajectory planning to enable adaptive terrain traversal and smooth motion.
2. **Decoupling Capacitor Improvement:** Add 100 μF ceramic or electrolytic capacitors at individual servo supply lines to suppress high-frequency transient current spikes during rapid joint acceleration and further reduce PWM noise coupling to the logic domain.
3. **Dual-Camera Chassis Integration:** Mount the RGB and NoIR camera modules in a fixed, co-aligned orientation on the robot body frame and validate synchronized dual-camera image capture during active robot motion.
4. **ORB Image Registration Deployment:** Implement and benchmark the ORB feature-matching and affine homography pipeline on the Raspberry Pi 4 to achieve sub-pixel spatial alignment between RGB and NIR frames.
5. **Multi-Index Pipeline Validation:** Deploy and validate the full NDVI, GNDVI, and SAVI computation pipeline using controlled spectral calibration targets of known reflectance to confirm end-to-end pipeline accuracy for all three indices before field deployment.
6. **ML Model Training and Validation:** Curate a labelled crop disease dataset, fine-tune MobileNetV2 with 4-channel (RGB + NDVI) input, convert to TFLite int8 format, and benchmark classification accuracy and inference latency on Raspberry Pi 4.
7. **Full System Integration Testing:** Conduct integrated locomotion, sensing, and inference testing in a controlled simulated field environment, followed by real-world agricultural field validation.

References

- [1] T. B. Shahi, C.-Y. Xu, A. Neupane, and W. Guo, “Recent advances in crop disease detection using UAV and deep learning techniques,” *Remote Sensing*, vol. 15, no. 9, p. 2450, 2023. DOI: 10.3390/rs15092450.
- [2] A. Nițu, C. Florea, M. Ivanovici, and A. Racoviteanu, “NDVI and beyond: Vegetation indices as features for crop recognition and segmentation in hyperspectral data,” *Sensors*, vol. 25, no. 12, p. 3817, 2025. DOI: 10.3390/s25123817.
- [3] C. J. Tucker, “Red and photographic infrared linear combinations for monitoring vegetation,” *Remote Sensing of Environment*, vol. 8, no. 2, pp. 127–150, 1979. DOI: 10.1016/0034-4257(79)90013-0.
- [4] A. A. Gitelson, Y. J. Kaufman, and M. N. Merzlyak, “Use of a green channel in remote sensing of global vegetation from EOS-MODIS,” *Remote Sensing of Environment*, vol. 58, no. 3, pp. 289–298, 1996. DOI: 10.1016/S0034-4257(96)00072-7.
- [5] A. R. Huete, “A soil-adjusted vegetation index (SAVI),” *Remote Sensing of Environment*, vol. 25, no. 3, pp. 295–309, 1988. DOI: 10.1016/0034-4257(88)90106-X.
- [6] A. Shirley Edward et al., “MLVI-CNN: A hyperspectral stress detection framework using machine learning-optimized indices and deep learning for precision agriculture,” *Frontiers in Plant Science*, vol. 16, p. 1631928, 2025. DOI: 10.3389/fpls.2025.1631928.
- [7] J. Verrelst et al., “Optical remote sensing and the retrieval of terrestrial vegetation biogeophysical properties—A review,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 108, pp. 273–290, 2015. DOI: 10.1016/j.isprsjprs.2015.05.005.
- [8] A. Upadhyay et al., “Deep learning and computer vision in plant disease detection: A comprehensive review of techniques, models, and trends in precision agriculture,” *Artificial Intelligence Review*, vol. 58, no. 3, p. 92, 2025. DOI: 10.1007/s10462-024-11100-x.

- [9] M. Shafay et al., “Recent advances in plant disease detection: Challenges and opportunities,” *Plant Methods*, vol. 21, no. 1, p. 140, 2025. DOI: 10.1186/s13007-025-01450-0.
- [10] J. Qi, A. Chehbouni, A. R. Huete, Y. H. Kerr, and S. Sorooshian, “A modified soil adjusted vegetation index,” *Remote Sensing of Environment*, vol. 48, no. 2, pp. 119–126, 1994. DOI: 10.1016/0034-4257(94)90134-1.
- [11] Y. Bai et al., “Hyperspectral remote sensing for monitoring crop disease: Applications, challenges, and perspectives,” *IEEE Geoscience and Remote Sensing Magazine*, 2025. DOI: 10.1109/MGRS.2024.3507068.
- [12] X. Zhang, B. A. Vinatzer, and S. Li, “Hyperspectral imaging analysis for early detection of tomato bacterial leaf spot disease,” *Scientific Reports*, vol. 14, no. 1, p. 27 666, 2024. DOI: 10.1038/s41598-024-78650-6.
- [13] Y. E. García-Vera, A. Polochè-Arango, C. A. Mendiavelso-Fajardo, and F. J. Gutiérrez-Bernal, “Hyperspectral image analysis and machine learning techniques for crop disease detection and identification: A review,” *Sustainability*, vol. 16, no. 14, p. 6064, 2024. DOI: 10.3390/su16146064.
- [14] L. Chen, Y. Wu, N. Yang, and Z. Sun, “Advances in hyperspectral and diffraction imaging for agricultural applications,” *Agriculture*, vol. 15, no. 16, p. 1775, 2025. DOI: 10.3390/agriculture15161775.
- [15] M. Reis Pereira et al., “Plant disease diagnosis based on hyperspectral sensing: Comparative analysis of parametric spectral vegetation indices and nonparametric Gaussian process classification approaches,” *Agronomy*, vol. 14, no. 3, p. 493, 2024. DOI: 10.3390/agronomy14030493.